```
import torch
import torchvision

from torchvision import transforms
from datasets import load_dataset
from torchvision.utils import make_grid
import matplotlib.pyplot as plt

import torch.nn.functional as F
from diffusers import DDPMScheduler
from diffusers import UNet2DModel
from datasets import load_dataset
from PIL import Image
```

```
dataset= load_dataset("huggan/smithsonian_butterflies_subset",split='train')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:103: UserWarning:
Error while fetching `HF_TOKEN` secret value from your vault: 'Requesting secret HF_TOKEN timed out. Secrets can only be fet
You are not authenticated with the Hugging Face Hub in this notebook.
If the error persists, please let us know by opening an issue on GitHub (https://github.com/huggingface/huggingface_hub/issu
  warnings.warn(
```

README.md: 100% 609/609 [00:00<00:00, 56.3kB/s]

```
Repo card metadata block was not found. Setting CardData to empty.
WARNING:huggingface_hub.repocard:Repo card metadata block was not found. Setting CardData to empty.
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
```

dataset_infos.json: 1.65k/? [00:00<00:00, 131kB/s]

data/train-00000-of-00001.parquet: 100% 237M/237M [00:02<00:00, 294MB/s]

Generating train split: 100% 1000/1000 [00:03<00:00, 319.27 examples/s]

```
image_size=64

preprocess=transforms.Compose([
        transforms.Resize((image_size, image_size)),  # Resize
        transforms.RandomHorizontalFlip(),  # Randomly flip (data augmentation)
        transforms.ToTensor(),  # Convert to tensor (0, 1)
        transforms.Normalize([0.5], [0.5]),  # Map to (-1, 1)
    ])
```

```
dataset.set_transform(preprocess)
batch_size=64
def transform(examples):
    images = [preprocess(image.convert("RGB")) for image in examples["image"]]
    return {"images": images}

dataset.set_transform(transform)

train_dataloader = torch.utils.data.DataLoader(
    dataset, batch_size=batch_size, shuffle=True
)

device="cuda" if torch.cuda.is_available() else "cpu"
```

```
def show_images(image):
    plt.imshow(make_grid(image*0.5+0.5).permute(1,2,0))
```

```
batch=next(iter(train_dataloader))
show_images(batch['images'][:8])
```

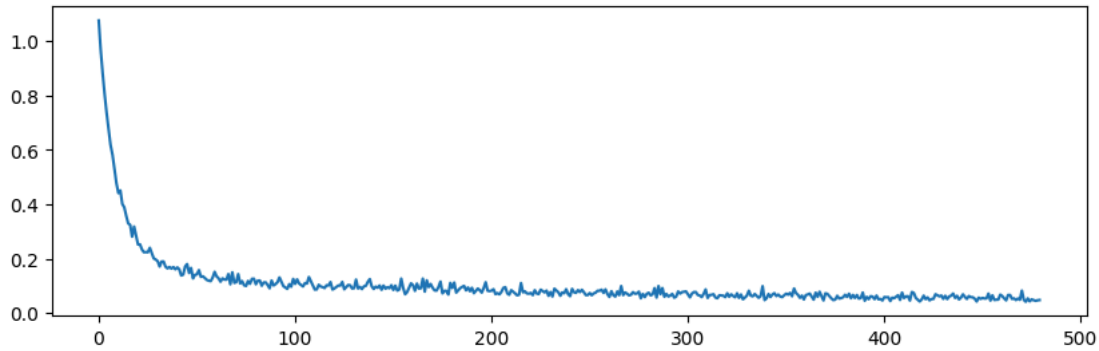
```
scheduler = DDPMScheduler(num_train_timesteps=1000, beta_start=0.001, beta_end=0.02)
model=UNet2DModel(sample_size=64,block_out_channels=(64,128,256,512,512,512,512),
                  down_block_types=('DownBlock2D', 'AttnDownBlock2D', 'AttnDownBlock2D',\
                                    'AttnDownBlock2D','AttnDownBlock2D','AttnDownBlock2D',\
                                    'AttnDownBlock2D'),
                   up_block_types=('AttnUpBlock2D','AttnUpBlock2D','AttnUpBlock2D',\
                                   'AttnUpBlock2D', 'AttnUpBlock2D', 'AttnUpBlock2D',\
                                   'UpBlock2D'))
```

```
num_epochs=30
ls=1e-4
model=model.to(device)
optimizer = torch.optim.AdamW(model.parameters(), lr=ls)
losses = []

for epoch in range(num_epochs):
    for step, batch in enumerate(train_dataloader):

        clean_images = batch["images"].to(device)

        noise = torch.randn(clean_images.shape).to(clean_images.device)
        timesteps = torch.randint(
            0,
            scheduler.num_train_timesteps,
            (clean_images.shape[0],),
            device=clean_images.device,
        ).long()

        noisy_images = scheduler.add_noise(clean_images, noise, timesteps)

        noise_pred = model(noisy_images, timesteps, return_dict=False)[0]

        loss = F.mse_loss(noise_pred, noise)

        losses.append(loss.item())
        loss.backward(loss)
        optimizer.step()
        optimizer.zero_grad()
    if (epoch+1)%10==0:
        print(f"Epoch {epoch+1} Loss {loss}")
```

```
/usr/local/lib/python3.12/dist-packages/diffusers/configuration_utils.py:173: FutureWarning: Accessing config attribute `num
  deprecate("direct config name access", "1.0.0", deprecation_message, standard_warn=False)
Epoch 10 Loss 0.10868462920188904
Epoch 20 Loss 0.058014847338199615
Epoch 30 Loss 0.04743536561727524
```

```
plt.figure(figsize=(10,3))
plt.plot(losses)
```

```
[<matplotlib.lines.Line2D at 0x7962ea4a4b00>]
```




```
from diffusers import DDPMPipeline

pipe= DDPMPipeline(unet=model,scheduler=scheduler)
pipe.to(device)
```

```
DDPMPipeline {
  "_class_name": "DDPMPipeline",
  "_diffusers_version": "0.37.1",
  "scheduler": [
    "diffusers",
    "DDPMScheduler"
  ],
  "unet": [
    "diffusers",
    "UNet2DModel"
  ]
}
```

```
pipe().images[0]
```

```
100%                                                      1000/1000 [00:52<00:00, 18.38it/s]
```

```
pipe().images[0]
```

```
100%                                                      1000/1000 [00:52<00:00, 20.08it/s]
```